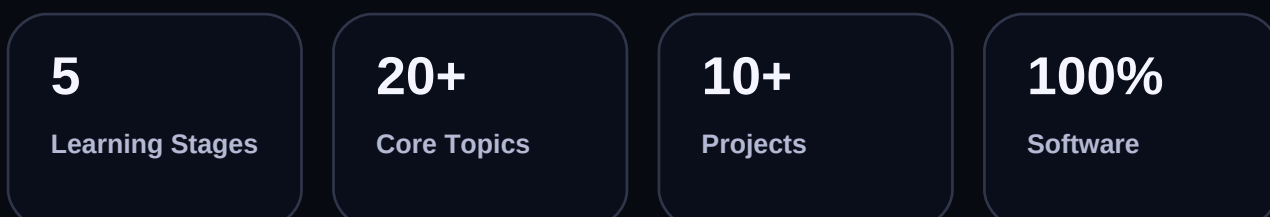
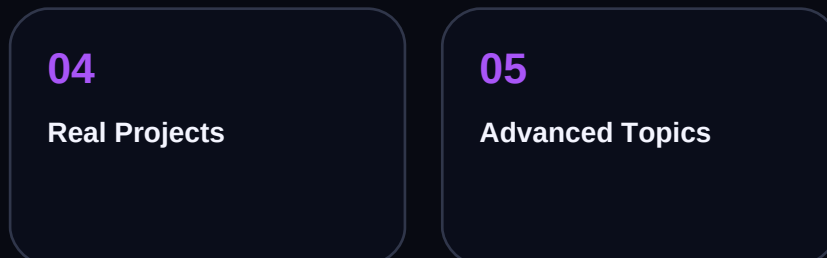
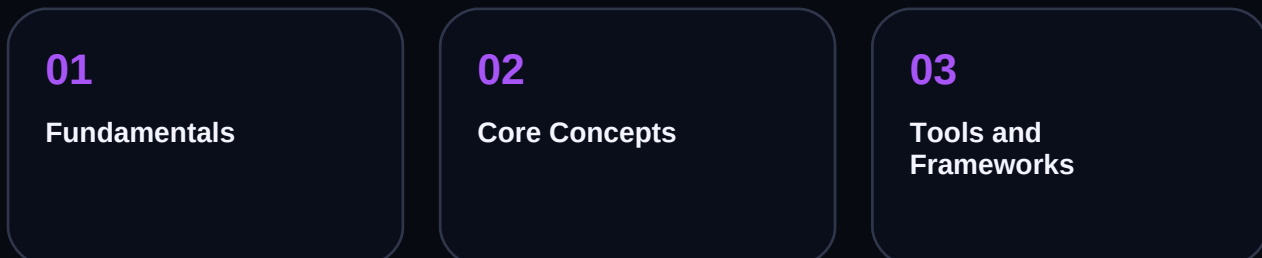
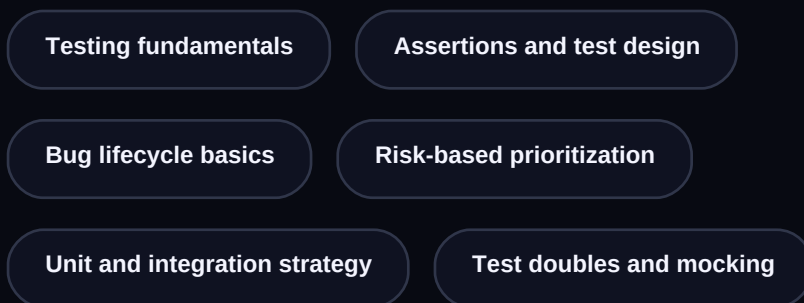




SOFTWARE TESTING

Complete Learning Roadmap

Establish quality confidence with robust unit, integration, and end-to-end testing strategy.



Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Testing fundamentals •
Assertions and test design •
Bug lifecycle basics •
Risk-based prioritization

02

Core Concepts

Unit and integration strategy •
Test doubles and mocking •
Regression planning •
Exploratory testing

03

Tools and Frameworks

Jest/Vitest •
Playwright/Cypress

04

Real Projects

Critical path automation suite •
Flaky test remediation •
Performance test baseline •
Release gate checks

05

Advanced Topics

Quality metrics governance •
Shift-left strategy • Reliability
engineering collaboration •
Test architecture at scale

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.



Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Testing fundamentals
- Assertions and test design
- Bug lifecycle basics
- Risk-based prioritization

Tools & Frameworks

- Testing fundamentals
- Assertions and test design
- Bug lifecycle basics
- Risk-based prioritization

Recommended Projects

- Software Testing workshop using Testing fundamentals
- Software Testing portfolio project with Assertions and test design

Stage 1 Milestone Complete Software Testing workshop using Testing fundamentals and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Unit and integration strategy
- Test doubles and mocking
- Regression planning
- Exploratory testing

Tools & Frameworks

- Unit and integration strategy
- Test doubles and mocking
- Regression planning
- Exploratory testing

Recommended Projects

- Software Testing workshop using Unit and integration strategy
- Software Testing portfolio project with Test doubles and mocking

Stage 2 Milestone Complete Software Testing workshop using Unit and integration strategy and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- Jest/Vitest
- Playwright/Cypress
- API testing tools
- CI test orchestration

Tools & Frameworks

- Jest/Vitest
- Playwright/Cypress

Recommended Projects

- Software Testing workshop using Jest/Vitest
- Software Testing portfolio project with Playwright/Cypress

Stage 3 Milestone Complete Software Testing workshop using Jest/Vitest and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Critical path automation suite
- Flaky test remediation
- Performance test baseline
- Release gate checks

Tools & Frameworks

- Critical path automation suite
- Flaky test remediation
- Performance test baseline
- Release gate checks

Recommended Projects

- Software Testing workshop using Critical path automation suite
- Software Testing portfolio project with Flaky test remediation

Stage 4 Milestone Complete Software Testing workshop using Critical path automation suite and document the decisions, trade-offs, and validation checks before moving to Stage 5.



Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

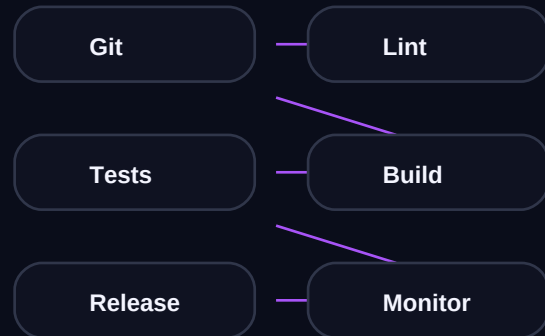
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Quality metrics governance
- Shift-left strategy
- Reliability engineering collaboration
- Test architecture at scale

CI/CD Pipeline



Stage 5 Milestone

A production-grade software testing workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

Software Testing docs and community guides

Link: <https://testing-library.com/docs/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/qa>

PAID

Structured specialization track focused on Software Testing

Link: <https://roadmap.sh/qa>

FREE

Software Testing roadmap reference

Link: <https://roadmap.sh/qa>

FREE

Official documentation

Link: <https://testing-library.com/docs/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Software Testing?
- Which project best proves Stage 4 competency in Software Testing?
- What trade-offs appear in Stage 5 decisions for Software Testing?